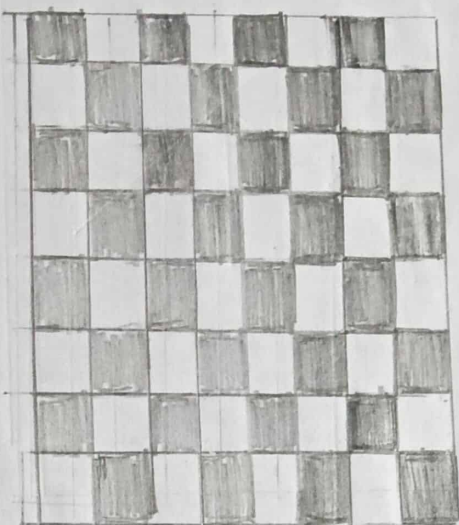


Index

S. No.	Name of the Experiment	Page No.	Date of Experiment	Date of Submission	Remarks
01.	Create a Chess Board	1	02/3/2026	5/3/26	
02.	Draw a Line using DDA	2	5/3/2026	5/3/26	
03.	Draw a Line using Bresenham's algorithm	3	9/3/2026	11/3/26	
04.	Draw a Circle	4	11/3/2026	11/3/26	
05.	Draw a Rectangle	5	16/3/2026	18/3/26	
06.	Draw a Ellipse	6	18/3/2026	18/3/26	
07.	Draw a Various shapes	7	23/3/26	25/3/26	
08.	Move a Car	8	25/3/26	25/3/26	
09.	Create smiling face animation	9	1/4/26	6/4/26	
10.	Draw a Pie-chart	10	6/4/26	6/4/26	
11.	Draw a Bezier surface	11-12	8/4/26	13/4/26	
12.	Display circles in a circle	13	13/4/26	13/4/26	
13.	Demonstrate Countdown	14	22/4/26	27/4/26	
14.	Demonstrate Traffic light Program	15-16	27/4/26	27/4/26	
15.	Restrict Mouse pointer in a circle.	17-18	6/5/26	20/5/26,	

Output



1. Create a chess board

```

#include<graphics.h>
#include<conio.h>
int main()
{
    int gd = DETECT, gm;
    int x=50, y=50;
    int size = 50;
    initgraph(&gd, &gm, "");
    for (int row=0; row<8; row++)
    {
        for (int col=0; col<8; col++)
        {
            if (row+col)%2==0)
                setfillstyle(SOLID_FILL, WHITE);
            else
                setfillstyle(SOLID_FILL, BLACK);
            bar(x+col*size,
                y+row*size,
                x+(col+1)*size,
                y+(row+1)*size);
        }
    }
    getch();
    closegraph();
    return 0;
}

```

2. Draw line using BDA

```

#include <graphics.h>
#include <conio.h>
#include <math.h>

int main() {
    int gd = Detect, gm, x1, y1, x2, y2, dx, dy, steps, i;
    float xinc, yinc;
    initgraph(&gd, &gm, "");
    printf("Enter x1 and y1\n");
    scanf("%d %d", &x1, &y1);
    printf("Enter x2 and y2\n");
    scanf("%d %d", &x2, &y2);
    dx = x2 - x1;
    dy = y2 - y1;
    if (abs(dx) >= abs(dy))
        steps = dx;
    else
        steps = dy;
    xinc = dx / steps;
    yinc = dy / steps;
    for (i = 0; i < steps; i++) {
        pixelround(x1, round(y1), 15);
        x1 = x1 + xinc;
        y1 = y1 + yinc;
        delay(100);
    }
    getch();
    closegraph();
    return 0;
}

```

Output.

Enter x1 and y1 800 800
Enter x2 and y2 300 300

3 Draw a line using Bresenham's algorithm

#include <graphics.h>

#include <conio.h>

#include <stdio.h>

int main() {

int gd = DETECT, gm, x1, y1, x2, y2, dx, dy, r, p;

initgraph(&gd, &gm, "");

printf("Enter x1 and y1\n");

scanf("%d %d", &x1, &y1);

printf("Enter x2 and y2\n");

scanf("%d %d", &x2, &y2);

dx = x2 - x1;

dy = y2 - y1;

p = 2 * dy - dx;

putpixel(x1, y1, 15);

for (f = 0; f < dx; f++) {

if (p < 0) {

x1++

p = p + 2 * dy;

}

else {

x1++

y1++;

p = p + 2 * dy - 2 * dx;

}

putpixel(x1, y1, 15);

delay(100);

}

getch();

closegraph();

return 0; }

4 Draw a Circle.

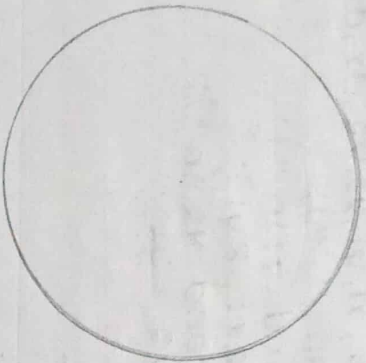
```

#include<graphics.h>
#include<conio.h>

int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");
    circle(50, 50, 10);
    getch();
    closegraph();
    return 0;
}

```

Output



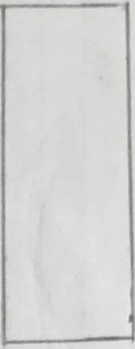
05. Draw a Rectangle

```

#include<graphics.h>
#include<conio.h>
int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");
    rectangle(100, 100, 300, 200);
    getch();
    closegraph();
    return 0;
}

```

Output



06. Draw a ellipse

```

#include<graphics.h>
#include<conio.h>
int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");
    ellipse(100, 30, 0, 360, 60, 40);
    getch();
    closegraph();
    return 0;
}

```

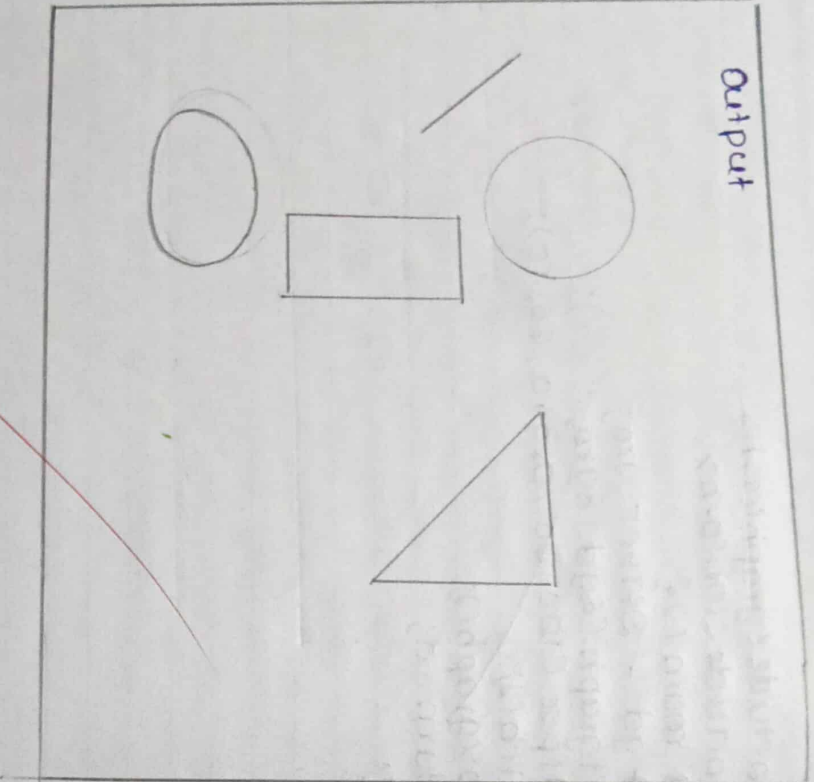
Output



07. Draw various shapes (line, rectangle, circle, ellipse, triangle)

```
#include <graphics.h>
#include <conio.h>
int main()
{
    int gd = DETECT, gm;
    intgraph(&gd, &gm, "");
    rectangle(100, 150, 180, 250);
    line(20, 120, 80, 150);
    circle(80, 60, 40);
    ellipse(100, 30, 0, 360, 60, 40);
    line(100, 100, 300, 300);
    line(300, 300, 300, 100);
    getch();
    closegraph();
    return 0;
}
```

Output



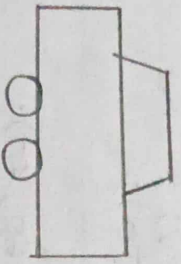
08. Move a Car

```

#include <graphics.h>
#include <iostream.h>
int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");
    for (int i = 0; i < 400; i++) {
        cleardevice();
        rectangle(50 + i, 200, 150 + i, 250);
        line(50 + i, 200, 90 + i, 170);
        line(90 + i, 170, 120 + i, 170);
        line(120 + i, 170, 140 + i, 200);
        circle(75 + i, 250, 10);
        circle(125 + i, 250, 10);
        delay(5);
    }
    getch();
    closegraph();
    return 0;
}

```

Output



09 Create Smiling face animation

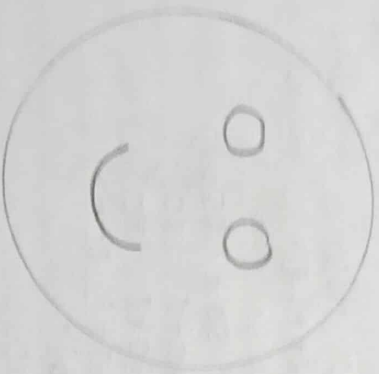
```

#include <graphics.h>
#include <conio.h>

int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");
    circle(300, 200, 60);
    ellipse(325, 175, 0, 360, 7, 5);
    ellipse(375, 175, 0, 360, 7, 5);
    setfillstyle(1, YELLOW);
    floodfill(300, 160, WHITE);
    setcolor(BLACK);
    ellipse(300, 210, 205, 335, 20, 9);
    ellipse(300, 210, 205, 335, 80, 10);
    ellipse(300, 210, 205, 335, 80, 11);
    getch();
    closegraph();
    return 0;
}

```

Output



10. Draw a Pie-chart

```
#include <graphics.h>
#include <conio.h>
int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");
```

```
setfillstyle(C1, RED);
outtextxy(10, 20, "C-40%");
pieslice(C150, 150, 0, 150, 40);
```

```
setfillstyle(C1, BLUE);
outtextxy(10, 40, "C++-30%");
pieslice(C150, 150, 150, 150, 200, 40);
```

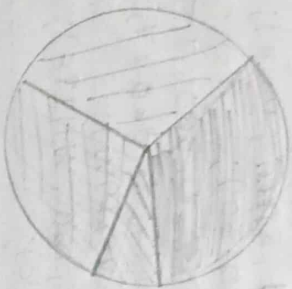
```
setfillstyle(C1, YELLOW);
outtextxy(10, 60, "JAVA-20%");
pieslice(C150, 150, 200, 250, 40);
```

```
setfillstyle(C1, GREEN);
outtextxy(10, 80, "PYTHON-10%");
pieslice(C150, 150, 250, 360, 40);
```

```
getch();
closegraph();
return 0;
}
```

Output

C-40%
C++-30%
JAVA-20%
PYTHON-10%



11. Draw a Bezier Surface.

```
#include <graphics.h>
#include <iostream.h>
#include <math.h>
#define STEPS 100
void bezierCurve(int x[4], int y[4])
```

```
{
    float t;
    for (t=0; t<=1; t+=1.0/STEPS)
```

```
{
    float xt = pow(1-t, 3) * x[0] +
                3 * t * pow(1-t, 2) * x[1] +
                3 * t * t * (1-t) * x[2] +
                pow(t, 3) * x[3];
```

```
float yt = pow(1-t, 3) * y[0] +
            3 * t * pow(1-t, 2) * y[1] +
            3 * t * t * (1-t) * y[2] +
            pow(t, 3) * y[3];
```

```
putpixel(xt, yt, WHITE);
```

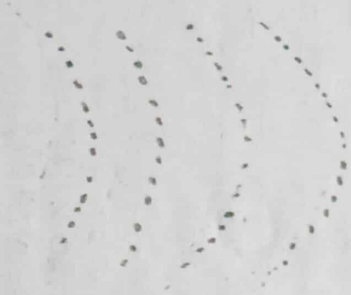
```
}
```

```
{
```

```
int gd = DETECT, gm;
initgraph(&gd, &gm, "");
int px[4][4] = {
```

```
{100, 150, 200, 250,
```

Output



```

x[100, 150, 800, 250],
x[100, 150, 800, 250],
x[100, 150, 800, 250],
y;

```

```

int py[4][4] = {

```

```

    {100, 50, 50, 100},
    {150, 100, 100, 150},
    {800, 150, 150, 800},
    {250, 800, 800, 250},

```

```

};

```

```

// Draw Curve joining surface

```

```

for (int i=0; i<4; i++)

```

```

{

```

```

    int x[4], y[4];

```

```

    for (int j=0; j<4; j++)

```

```

    {

```

```

        x[j] = px[i][j];

```

```

        y[j] = py[i][j];

```

```

    }

```

```

    bezierCurve(x, y);

```

```

}

```

```

getch();

```

```

closegraph();

```

```

return 0;

```

```

}

```

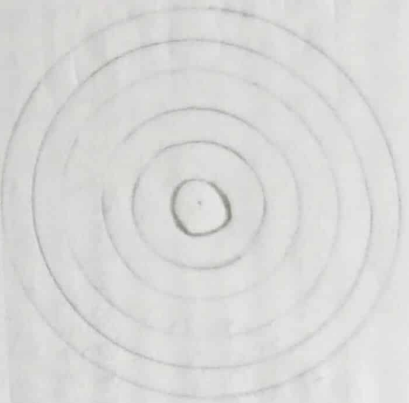
12. Display Circles in Circle.

```

#include <graphics.h>
#include <conio.h>
int main() {
    int gd = DETECT, gm, i;
    initgraph(&gd, &gm, "");
    for (i = 20; i <= 180; i = i + 20) {
        circle(300, 150, i);
        getch();
        closegraph();
        return 0;
    }
}

```

Output



Output

5
4
3
2
1
0
START!

Expt. No. 13

Date 22/4/26
Page No. 14

13. Demonstrate Countdown

```
#include <graphics.h>
#include <conio.h>
#include <dos.h>
#include <stdio.h>

int main() {
    int gd = DETECT, gm;
    char num[5];

    int i;
    initgraph(&gd, &gm, "");
    settextstyle(DEFAULT_FONT, HORIZ, BIR, 5);
    for (i = 5; i >= 0; i--) {
        clrdevice();
        sprintf(num, "%d", i);
        outtextxy(300, 200, num);
        delay(1000);
    }
    outtextxy(350, 300, "START!");
    getch();
    closegraph();
    return 0;
}
```

Teacher's Signature

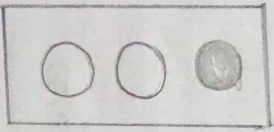
14. Demonstrate Traffic Light Program

```

#include <graphics.h>
#include <conio.h>
int main() {
    int gd = DETECT, gm;
    while (cibhst()) {
        rectangle(20, 20, 100, 200);
        setfillstyle(1, RED);
        circle(60, 50, 20);
        floodfill(60, 50, WHITE);
        circle(60, 100, 20);
        circle(60, 150, 20);
        delay(2000);
        cleardevice();
        rectangle(20, 20, 100, 200);
        circle(60, 50, 20);
        setfillstyle(1, YELLOW);
        circle(60, 100, 20);
        floodfill(60, 100, WHITE);
        circle(60, 150, 20);
        delay(2000);
        cleardevice();
        rectangle(20, 20, 100, 200);
        circle(60, 50, 20);
        setfillstyle(1, GREEN);
        circle(60, 150, 20);
    }
}

```

Output



```

1100d111 (60,150,WHITE);
delay (2000);
clearDeviceC();
? getch();
closegraphC();
return 0;
}
    
```

Teacher's Signature _____

15. Restrict Mouse Pointer on a circle.

```

#include <graph.h>
#include <math.h>
#include <conio.h>
int main() {
    int x, y;
    int xc = 300, yc = 200, r = 100;
    initwindow(640, 480, "Mouse Restriction");
    circle(xc, yc, r);
    while (kbhit()) {
        int mx = mousex();
        int my = mousey();
        char device();
        circle(xc, yc, r);
        int dist = sqrt((xc - mx) * (xc - mx) +
            (yc - my) * (yc - my));
        if (dist < r) {
            setcolor(CRED);
            circle(x, y, 3);
        }
        else {
            setcolor(CRGB);
            circle(xc, yc, 3);
            delay(200);
        }
    }
}

```

Output



Expt No. _____

Date _____

Page No. 18

closegraph (i);
return 0;

Teacher's Signature

[Signature]
20/8/20

Index

S. No.	Name of the Experiment	Page No.	Date of Experiment	Date of Submission	Remarks
1.	Write a JavaScript program to check whether the given number is perfect, abundant or deficient use alert box to display the result.	1	03/3/26	9/3/26	
2.	Design a JavaScript program to display the multiplication table expecting the number and limit.	2	09/3/26	9/3/26	
3.	Write a JavaScript program to store different colors in an Array and change the background color of the page using these array elements	3	12/3/26	10/3/26	
4.	Write a JavaScript function program to change the text color and back color of textbox using onfocus and onblur events	4	17/3/26	22/3/26	
5.	Write a Javascript function to display current Date and Time	5	27/3/26	13/4/26	
6.	Write a Javascript program that calculates the squares and cubes	6	13/4/26	13/4/26	

Index

S. No.	Name of the Experiment	Page No.	Date of Experiment	Date of Submission	Remarks
	of a numbers 0 to 10 and output HTML text that displays the resulting values in HTML table format.				
07.	Write a Javascript program to calculate factorial of any number by using HTML	7	22/4/26	17/5/26	
08	Write a JavaScript function to check whether a given value is ±p value or not	8	17/5/26	17/5/26	
09.	Design a student registration form and apply validation on it by using external JavaScript	9-11	14/5/26	18/5/26	
10.	Design a Simple calculator by using HTML, CSS and JavaScript	12-14	18/5/26	18/5/26	

Chandru

Output

Enter a number 10

10 is a deficient number

Enter a number 0

0 is a invalid number

01. Write a JavaScript program to check whether the given number is perfect, abundant or deficient use alert box to display the output

```

<doctype html>
<html>
  <head>
    <title> Number check </title>
  </head>
  <body>
    <script>
      var num=parseFloat(prompt("enter a number:"));
      var sum=0;

      for (var i=1; i<num; i++) {
        if (num%i==0) {
          sum=sum+i;
        }
      }

      if (num==0) {
        alert (num+" is a invalid number");
      }
      else if (sum==num) {
        alert (num+" is a perfect number");
      }
      else if (sum>num) {
        alert (num+" is an abundant number");
      }
      else {
        alert (num+" is a deficient number");
      }
    </script>
  </body>
</html>

```

02. Design a Javascript program to display the multiplication table by expecting the number and limit

```
<!doctype html>
```

```
<html>
```

```
<head>
```

```
<title> Multiplication Table</title>
```

```
</head>
```

```
<body>
```

```
<script>
```

```
var num = prompt("enter a number");
```

```
var limit = prompt("enter the limit");
```

```
if (num > 0 && limit > 0) {
```

```
for (var i = 1; i <= limit; i++) {
```

```
document.write (num + "x" + i + " = " + (num * i)
```

```
 + "<br>");
```

```
}
```

```
else {
```

```
document.write("Please enter positive
```

```
integers only");
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Output

enter a number 3

enter limit 10

3x1 = 3

3x2 = 6

3x3 = 9

3x4 = 12

3x5 = 15

3x6 = 18

3x7 = 21

3x8 = 24

3x9 = 27

3x10 = 30

Output

change 01 Background color

change color

03. Write a JavaScript program to store different colors in an array and change the background color of the page using these array elements

```

<doctype html>
<html>
  <head>
    <title> Change Background color </title>
  </head>
  <script>
    var colors = ["red", "blue", "green",
    "yellow", "pink"];
    var index = 0;
    function changeColor() {
      document.body.style.backgroundColor =
        colors[index];
      index++;
    }
    (index == colors.length) {
      index = 0;
    }
  </script>
  <body>
    <h2> Change 01 Background Colors </h2>
    <button onclick = "changeColor()"> Change
    color </button>
  </body>
</html>

```

04 Write a JavaScript program to change the text color and back color of a text box using onfocus and onblur events

```

<!DOCTYPE html>
<html>
<body>
<h3> Changing the Text and Back color of
    Text Box </h3>
    Enter a Text:
    <input type="text" id="box">
    <script>
        var x = document.getElementById("box");
        x.onfocus = function() {
            x.style.backgroundColor = "yellow";
            x.style.color = "red";
        }
        x.onblur = function() {
            x.style.backgroundColor = "pink";
            x.style.color = "black";
        }
    </script>
</body>
</html>

```

Output.

Changing the text and back color of the
Text box

Enter name : Akash

05. Write a JavaScript Function to display current date and Time

```

<!DOCTYPE html>
<html>
  <head>
    <title> Current Date and Time </title>
  <body>
    <h2> Current Date and Time </h2>
    <p id = "demo"> </p>
    <script>
      var today = new Date();
      document.getElementById("demo").innerHTML =
        today;
    </script>
  </body>
</html>

```

Output

Current Date and Time
Wed Apr 01 2026 11:55:46 GMT +
0530 Cindia Standard Time)

Output
Squares and Cubes (0 to 10)

Number	Squares	Cubes
0	0	0
1	1	1
2	4	8
3	9	27
4	16	64
5	25	125
6	36	216
7	49	343
8	64	512
9	81	729
10	100	1000

Q6. Write a Javascript program that calculates the squares and cubes of the numbers 0 to 10 and outputs html text that displays the resulting values in a HTML table format

```

<html>
<body>
  <h3> Squares and Cubes (0 to 10) </h3>
  <table border = "1">
    <tr>
      <th> Number </th>
      <th> Squares </th>
      <th> Cubes </th>
    </tr>
    <script>
      for (let i = 0; i <= 10; i++) {
        document.write ("<tr>");
        document.write ("<td>" + i + "</td>");
        document.write ("<td>" + (i*i) + "</td>");
        document.write ("<td>" + (i*i*i) + "</td>");
        document.write ("</tr>");
      }
    </script>
  </table>
</body>
</html>

```

Output

Enter Number

5

120

Find factorial

07. Write a JavaScript program to calculate factorial of any number by using HTML.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title> Factorial of a Number </title>
```

```
</head>
```

```
<body>
```

```
<input type="number" id="n" placeholder="Enter number">
```

```
<button onclick="fact()"> Find Factorial
```

```
</button>
```

```
<p id="ans"> </p>
```

```
<script>
```

```
function fact() {
```

```
let n=document.getElementById("n").value;
```

```
let f=1;
```

```
for (let i=1; i<=n; i++) {
```

```
    f=i*f;
```

```
}
```

```
document.getElementById("ans").innerHTML=f;
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Output

IP address checker

1.1.1.1

Checker

Valid IP address

Q8. Write a JavaScript function to check whether a given value is IP value or not

```

<!DOCTYPE html>
<html>
<body>
<h3>IP address checker</h3>
<input type="text" id="ip">
<button onclick="checkIP()"> Check</button>
<div id="result"></div>
</script>
</body>
</html>

function checkIP() {
    var ip = document.getElementById("ip").value;
    var parts = ip.split(".");
    if (parts.length == 4) {
        parts[0] = parseInt(parts[0]);
        parts[1] = parseInt(parts[1]);
        parts[2] = parseInt(parts[2]);
        parts[3] = parseInt(parts[3]);
        document.getElementById("result").innerHTML = "Valid IP Address";
    }
    else {
        document.getElementById("result").innerHTML = "Invalid IP Address";
    }
}
</script>
</body>
</html>

```

Output:

Student Registration Form

Name:

Email:

Gender: ☐ Male ☒ Female

Course:

Form Submitted successfully!

Expt. No. 09

Date 14/5/20

Page No. 9

09

Design a student registration form and apply validation on it by using external JavaScript.

Save as index.html

<!DOCTYPE html>

<html>

<head>

<title> Student Registration Form </title>

<script src = "validate.js"> </script>

</head>

<body>

<h2> Student Registration Form </h2>

<form name = "studentForm" onsubmit = "return

validateForm()">

Name:<input type = "text" name = "name">

Email : <input type = "text" name = "email">

Gender : <input type = "radio" name = "gender"

value = "male"> Male

<input type = "radio" name = "gender"

value = "female"> Female

Course :

<select name = "course">

<option value = " " > -- Select Course -- </option>

<option value = " " > B.Tech </option>

<option value = "BBA" > BBA </option>

<option value = "B.Tech" > B.Tech </option>

Teacher's Signature

```
</select>
<br> <br>
<input type = "submit" value = "SUBMIT">
</form>
</body>
</html>
```

validate.js

junction validateForm() {

let name = document.forms["studentForm"] ["name"]
value;

let email = document.forms["studentForm"] ["email"]
value;

let gender = document.forms["studentForm"] ["gender"];
let course = document.forms["studentForm"] ["course"]
value;

// Name validation

if (name == "") {

alert("Name must be filled out");
return false;
}

// email validation

let emailPattern = /^[a-zA-Z@[\]+\.[a-z]{2,3}\$/;

if (!emailPattern.match(email)) {
alert("Enter a valid email");
return false;
}

// gender validation

let gendersSelected = false;

for(let i=0; i < gender.length; i++) {

if (gender[i].checked) {

gendersSelected = true;

break;

}

}

if (genderSelected) {

alert("Please select gender");

return false;

}

// Course validation

if (course == "") {

alert("Please select a course");

return false;

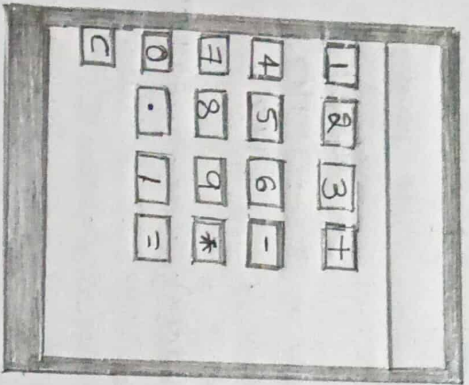
}

alert("Form submitted successfully");

return true;

}

Output



Expt No. 10.

Date 18/5/26
Page No. 18

10. Design a simple calculator by using html, css and JavaScript

Save as -index.html

<!DOCTYPE html>

<html>

<head>

<title> Calculator</title>

<style>

body {

display: flex;

justify-content: center;

align-items: center;

height: 50vh;

}

• box {

border: 5px solid black;

}

</style>

<script src = "script.js"></script>

</head>

</body>

<div class = "box">

<input type = "text" id = "box" >

<button onclick = "show('1')> 1 </button>

<button onclick = "show('2')> 2 </button>

<button onclick = "show('3')> 3 </button>

<button onclick = "show('+'> + </button>

Teacher's Signature

```
<br> <br>
<button onclick = "show ('4')">4 </button>
<button onclick = "show ('5')">5 </button>
<button onclick = "show ('6')">6 </button>
<button onclick = "show ('-1')">- </button>

<br> <br>

<button onclick = "show ('7')">7 </button>
<button onclick = "show ('8')">8 </button>
<button onclick = "show ('9')">9 </button>
<button onclick = "show ('*')">* </button>

<br> <br>

<button onclick = "show ('0')">0 </button>
<button onclick = "show ('.')">. </button>
<button onclick = "answer()">= </button>

<br> <br>

<button onclick = "clearBox()">C </button>

<div class = "box">
  <body>
    <html>

    Save as script.js

    function showCvalue()
    {
      document.getElementById("box").value += value;
    }

    function answer() {
      let x = document.getElementById("box").value;
```

Expt. No. _____

Date _____

Page No. 14

```
document.getElementById("box").value = eval("1");  
function clearBox()  
{  
    document.getElementById("box").value = " ";  
}
```

~~Seen~~
20/5/20

Teacher's Signature _____

```
document.getElementById("box").value = eval(x);
```

```
}
```

```
function clearBox()
```

```
{
```

```
document.getElementById("box").value = " ";
```

```
}
```

~~Seen~~
~~21/5/26~~